

Solar Monitor

- Solar Panel Fault Detection — interpretable first-pass inspection
- Team: YOUR TEAM NAME
- Track: AI

1. Problem

- Solar-panel abnormalities can be difficult to identify quickly.
- Operators need a clear indication of where to inspect.
- Manual inspection can require repeated visual checking and prioritization.

2. Solution

- Take a solar-panel image as input.
- Divide the image into regions and calculate an anomaly score.
- Select the most suspicious region.
- Highlight it with a red circle and generate a detection report.

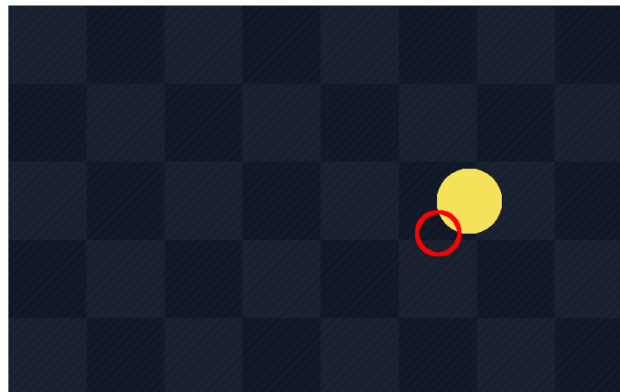
3. How It Works

- PPM image loader → region analysis → anomaly scoring → candidate region → annotation + report.
- Uses standard C++17 and does not require OpenCV.
- The marked region is a suspicious anomaly, not a confirmed physical fault.

4. Demo — Result

- The demo image contains a synthetic abnormal bright region.
- The detector identifies a candidate region and creates an annotated output.
- Three screenshots are included in demo/screenshots.

Solar Monitor — Result Output



Report

Solar Monitor Detection Report

=====

Image: /mnt/data/demo_input.ppm

Image size: 800 x 500

Candidate center: (550, 291)

Candidate radius: 27 pixels

Anomaly score: 64.8319

Grid cell: (5, 3)

5. IBM Technology Integration

- IBM Bob can serve as the conversational interface around the detector.
- A watsonx.ai layer can explain anomaly scores and summarize inspection results.
- IBM integration should be demonstrated as a genuine workflow, not only mentioned.

6. Impact & Future Scope

- Faster first-pass inspection and easier prioritization.
- Replace the heuristic with a trained visual/thermal anomaly model.
- Support multiple fault regions, confidence scores, historical monitoring and fleet analytics.
- Connect results to a conversational IBM Bob workflow.

7. Reproducibility

- Requires a C++17 compiler.
- Build both C++ source files.
- Generate demo_input.ppm, run the detector, then inspect output and report.
- Exact commands are documented in docs/setup-guide.md.